MILITARY SPECIFICATION

HEAD ASSEMBLY: 10541990
(PERISCOPE: M32C)

1. SCOPE

1.1 This specification covers a modular unit of the Tank Gunner's M32C Periscopes, herein referred to as "head assembly".

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATION

Military

MIL-O-13830	Optical Components for Fire Control Instruments; General Specification Govcrning the Manufacture, Assembly and Inspection of
MIL-F-13926	Fire Control Materiel; General Specification on Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-l09	Quality Assurance Terms and Definitions

DRAWINGS

US Army, Frankford Arsenal

F5549108	Telescope, Collimation (3X)
F8247391	Test Fixture for Periscope M32

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F8565556	Pressure Tester, Two-Station
F8575736	Vibration Adapter for Periscope M32
F1054O499	Universal Test Stand and Set-Up Instructions
F1054199O	Head Assembly
F10547434	10" Collimator
F10547987	Set-Up Instructions for Calibration Standards, Periscope M32
D10548766	Calibration Standard, 26° Elevation, Periscope M32
F10548767	Support Assembly Collimator
D11729723	Gage, Minimum Interchangeability
D117321OI	Gage, Maximum Interchangeability
D11732107	Gage, Maximum Interchangeability
D11732108	Gage, Minimum Interchangeability

PACKAGING DATA SHEET

10541990	Packaging of Head Assembly, 10541990

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMETNS

3.1 Fabrication. - The head assembly shall be manufactured in accordance with Drawing F1054199O and drawings pertaining thereto.

3.2 Interchangeability. - The head assembly cavities shall be capable of the insertion and removal of the gages specified in 4.6.12 without restriction.

3.3 General specification. - The contractor shall be responsible for adhereance to, and compliance with the following requirements of ?fIL-F-13926.

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Part identification and marking
e. Workmanship

3.4 Environmental.

3.4.1 Storage temperature. - Prior to meeting the requirements of 3.7 to 3.13 inclusive, the head assembly shall show no evidence of physical failure such as glass fracture or breakage, bonding or casting faulure due to storage at temperatures ranging from -80° to +160°F.

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3.4.2 Operating temperatures. - The head assembly shall meet the applicable requirements of 3.5 while exposed and thermally stabilized at ambient temperatures of -40° and +150°F.

3.5 Torque. - The running torque required to rotate the inout coupling through range specified in 3.7.4 shall not exceed 6.0 pound-inches (lb-ins) when checked at standard temperature. When checked at the temperature conditions specified in 3.4.2, the running torque shall not exceed 10.0 lb-ins.

3.6 Orientation. - The requirements of 3.7, 3.10 and 3.11 shall be met with the head assembly positioned as follows:

a. Datum Plane "Y" on a horizontal plane
b. Side "L" of the input coupling positioned 22 degrees 30 minutes with respect to a plane defined by the four mounting pads as measured in a vertical plane.

NOTE: The following references apply throughout this specification:

Datum Plane "W" - See Drawing F11727083

3.7 Collimation. - The requirements of 3.7.1 to 3.7.8 inclusive, shall be met with:

a. An incident line of sight which is originally perpendicular to a plane defined by the four mounting pads.
b. The input coupling positioned in accordance with 3.6.

3.7.1 Pre-vibration, Daylight viewing area.- The incident line of sight shall emerge perpendicular to Datum Plane "Y" in deflection within 0.20 mil and be elevated 20 mils + 0.30 mil "above" the deflection plane ad defined. The deflection direction is defined by the plane which is perpendicular to Datum Plane "Y" parallel to the intersection of Datum Planes "Y" & "W". The term "above" indicates that the emerging line of sight is rotated away from Datum Plane "W".

3.7.2 Pre-vibration, Infrared viewing area.- The incident line of sight shall emerge perpendicular to Datum Plane "Y" within 0.20 mil.

3.7.3 Unity power area.- The reflected line of sight shall be depressed 26 degrees below Datum Plane "Y" within 15 minutes, and be perpendicular to Datum Plane "W" within 15 minutes.

3.7.4 Excursion range.- Rotation of the input coupling shall cause the line of sight to deflect a minimum of 22.0 degrees above to 18.0 degrees below the horizontal plane.

3.7.4.1 Coupling travel.- Rotation of the input coupling shall be limited by contract with the coupling stop pin only.

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NOTE: At no time shall the input coupling be allowed to strike hard against the travel stops.

3.7.5 Plumb travel.- Starting with the line of sight in coincidence with a vertical target line at the horizontal position, the line of sight shall track: the target line within 0.25 mil when elevated and depressed through the range specified in 3.7.4.

3.7.6 Elevation accuracy.- This requirement shall be met with backlash removed and the angle measured in a vertical plane. The angle formed by the line of sight and side "L" of the input coupling shall remain comstant within 0.20 mil at any elevation setting from 18 degrees depression to 22 degrees elevation.

3.7.7 Backlash. - Backlash between the line of sight and the input coupling shall not exceed 0.05 mil.

3.7.8 Image tilt. - This requirement shall be met when viewing through either the daylight or infrared exit aperture. The image of a target line which is originally perpendicular to Datum Plane "Y" shall emerge perpendicular to Datum Plane "W" within 30 minutes of arc.

3.8 Resolution.

3.8.1 Daylight and infrared viewing areas,- Resolution shall be 6.0 seconds of arc or less when viewing through either the daylight or infrared exit apertures with a telescope having a 1.6 inch entrance pupil Resolution consists of detectable line structure and proper line count in all four meridians at one focus setting.

3.8.2 Unity power system. - Resolution shall be 45 seconds of arc or less when viewing through the unity-power exit aperture with an observation telescope of at least 1/2 inch entrance pupil.

3.9 Vibration. - The head assembly shall show no evidence of physical failure due to being vibrated in a vertical plane at a constant frequency of 30 cycles per second (cps) with an amplitude of 1/16 inch (1/8 inch total excursion) for a period of 5 minutes plus or minus 15 seconds.

3.10 Post-vibration, Daylight viewing area.- The incident line of sight shall remain within 0.10 mil of the collimation reading established in 3.7.1.

3.11 Post-vibration, Infrared viewing area.- The incident line of sight shall remain within 0.10 mil of the collimation reading established in 3.7.2.

3.12 Sealing. - The head assembly shall show no evidence of leakage in excess of 0.2 pounds per square inch (psi) when subjected to an internal pressure of 5.0 + 0.10 psi for a mininum of 2 hours, The interior atmosphere of the accepted head assembly shall contain nitrogen (having a dew

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point at least as low as minus 25° at atmospheric pressure) between 0.25 and 0.50) pound per square inch gage (psig).

3.13 Cleanliness.

a. Defects. - There shall be no evidence of moisture, grease, fingerprints-condensates, fractures, and adhesive separations on any glass component when viewing through either end.

b. Dirt. - There shall be no more than 3 particles of dirt or foreign material (classed as dirt) on any glass component. The size of any dirt particle shall not exceed 0.05.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements,

4.1.1 General provisions. - The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4,2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) head assemblies plus three (3) each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing. - When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, inclucling certificates of conformance for materials shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

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4.3 Inspection provisions.

4.3.1 Submission of product. - Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II and III herein. Examination and tests for-packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

4.3.2.2 100 Percent inspection. - The following requirements and associated tests shall be subjected to 100% inspection:

	CHARACTERISTIC	REQUIREMENTS	TEST PROCEDURE
1.	Interchangeability	3.2	4,6.12
2.	Resolution, daylight and infrared viewing areas	3.8.1	4.6.5
3.	Resolution, unity power system	3.8.2	4.6.6

NOTE: The tests in 4.3.2.2 shall be conducted at a temperature between +60° and +90°F.

TABLE I - CLASSIFICATION OF DEFECTS

CLASS / CHARACTERISTIC	REQUIREMENTS	TEST PROCEDURE
CRITICAL : NONE		
MAJOR: AQL 0.65% Defective		
101. Torque	3.5	4.6.3
102. Collimation, pre-vibration, daylight viewing area	3.7.1	4.6.4.1
103. Collimation, pre-vibration, infrared viewing area	3.7.2	4.6.4.2
104. Collimation, Unitv power area	3.7.3	4.6.4.3
105. Excursion range	3.7.4	4.6.4.5

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TABLE I - CLASSIFICATION OF DEFECTS (CON'T)

CLASS	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE			
MAJOR:	AQL 0.65% Defective		
106.	Coupling travel	3.7.4.1	4.6.4.6
107.	Plumb travel	3.7.5	4.6.4.7
108.	Elevation accuracy	3.7.6	4.6.4.8
109.	Backlash	3.7.7	4.6.4.4
110.	Image tilt	3.7.8	4.6.4.9
111.	Vibration	3.9	4.6.7
112.	Collimation, post-vibration, daylight viewing area	3.10	4.6.8
113.	Collimation, post-vibration, infrared viewing area	3.11	4.6.9
114.	Sealing	3.1.2	4.6.10
115.	Cleanliness	3.13	4.6.11
MINOR:	NONE		

NOTE: The tests of Table I shall. be conducted at a temperature between +60° and +90°F.

4.3.3 Acceptance and rejection. - Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-l05.

4.4 Special sampling.

4.4.1 General. - One head assembly shall be selected at random as a special sampling from each} 100 produced. The samples shall meet the requirements and tests in Table II.

TABLE II

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301 l	Fabrication	3.1	Applicable drawing - visual
302.	General specification	3.3	MIL-T-13926 - visual

NOTE: The examinations in Table 11 shall be conducted at a standard ambient temperature of between 60° and 90°F.

4.4.2 Environmental tests. - Three head assemblies shall be selected at random from each 50 produced, or from each month's production, whichever occurs first. Each sample shall meet the requirements and tests in Table III.

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TABLE III

	CHACTERISTIC	REQUIREMENTS	TEST PROCEDURES
303.	Storage temperatures (-80° and +160°F)	3.4.1	4.6.1
304 l	Torque, elevation input shaft (-40° and +150°F)	3.4.2, 3.5	4.6.2, 4.6.3

4.4.3 Failure of sample. - Should any item of a special sampling fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment. - Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- When the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design. - Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Inspection Documents when provided with the Technical Data Package (TDP).

4.5.2.2 Contractor design. - The contractor shall provide inspection equipment compatable with the “Test Methods and Procedures” specified in 4.6 of this specification and with the “Component Inspection” and “Test Facilities” requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment design requirements.

4.5.2.3.1 Sealing. - Pressure testing equipment to test sealing of the head assembly shall conform to the internal pressure test requirements of Drawing F8565556, Pressure Tester-Two Station.

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4.5.2.3.2 Performance. - Special test equipment to determine performance of the head assembly shall conform to the requirements of Drawing F8247391 (Test Fixture for Periscope M32 and Drawing F10540499 (Universal Test Stand and Set-Up Instructions). The "Test Stand Components" depicted on Drawing F1054O499 form a part of the test equipment.

4.6 Test methods and procedures.

NOTE: Caution shall be exercised during the following tests to prevent the input coupling from striking. hard against the travel stops.

4.6.1 Storage temperature. - The head assembly shall be exposed to the temperature conditions specified in 3.4.1, utilizing testing equipment in accordance with the "Test Facilities" requirements of MIL-F-13926. The head assembly shall be placed in the test chamber and the temperature of the chamber reduced gradually to -80°F. The head assembly shall remain at this temperature for a minimum of five hours. At the completion of this five hour period, the temperature shall then be gradually raised to +160°F. The head assembly shall remain at this temperature for a minimum of five hours. At the completion of this five hour period, the temperature of the test chamber shall be gradually reduced to room ambient temperature (+60° to +90°F) at which time the head assembly shall be removed from the test chamber. The head assembly shall then be subjected to a visual and tactile examination to determine that there is no evidence of physical failure, as depicted in 3.4.1. Caution shall be exercised during this test to aviod subjecting the head assembly to thermal shock.

NOTE: The rate of temperature change in the test chamber shall not exceed 20°F per hour throughout the temperature cycling tests of 4.6.1 and 4.6.2.

4.6.2 Extreme operating temperatures. - The head assembly shall be exposed to the temperature condition specified in 3.4.2, utilizing the testing equipment in accordance with the "Test Facilities" requirements of MIL-F-13926. The head assembly shall be placed in the test chamber and the temperature of the chamber reduced qradually to -40°F. The assembly shall remain at this temperature for a minimum of 5 hours. At the completion of this 5 hour period, and while at -40°F the head assembly shall be subjected to the tests in 4.6.3 to determine compliance with 3.4.2. upon completion of this portion of the test, the head assembly temperature shall be gradually raised to room ambient temperature. With the assembly in the test chamber the temperature shall be raised gradually to +150°F. The head assembly shall remain at this temperature for a period of 5 hours and at the conclusion. of this 5 hour period, and while at +150°F, the head assembly shall again be subjected to the tests in 4.6.3 to determine compliance with 3.4.2. Upon completing of this test the temperature of the test chamber shall be gradually reduced to room ambient temperature (60° to 90°F) at which tire the head assemnby shall be removed from the test chamber. Caution shall be exercised during this test to avoid subjecting the head assembly to thermal shock.

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4.6.3 Torque. - Standard torque measuring equipment capable of measuring the torque values specified in 3.5 shall be utilized for this test. An appropriate adapter to accommodate. the coupling shall be used with the torque measuring equipment. The running torque required to rotate the coupling through its full range of travel in each direction, shall conform to the applicable requirements of 3.5 for the temperatures indicated.

4.6.4 Performance. - To determine compliance with 3.6, the following tests shall be performed utilizing the special test equipment specified in 4.5.2.3.2. Set-up of the test equipment and installation of the head assembly under test shall be performed in accordance with the set-up and calibration instructions contained on Drawing F8247391. The Collimator Projector 10547434 shall be focused at 1200 meters. Unless otherwise specified herein, these tests shall be performed at the daylight viewing area (Head wedge 10541997).

4.6.4.1 Collimation, pre-vibration, daylight viewing, area. - Prior to performing this test and the following tests of 4.6.4.2 and 4.6.4.3, disengage the index pin "D" from the index plate “E” of the test fixture and place the 10 second level on surface “C” of the leveling base. The bubble of the level shall be centered within its graduation lines. If it is not, then adjustment is required in accordance with the procedures specified in paragraphs 7 and 8 of the set-up instructions contained on Drawing F8247391. With the index pin “D” disengaged from the index plate and surface “C” of the leveling base level, through the auto-collimator of the Support Assembly Collimator 10548767 positioned at the daylight viewing area and observe the projected target image. The center of the target image (apparent intersecting point of the horizontal and vertical reticle lines) leaving the penta prism shall be elevated 20 mils above the reference line of sight of the viewing auto-collimator within the tolerance specified in 3.7.1, when measured by means of the target image graduations and the auto-collimator’s micrometer knobs. Record this measurement for use in the subsequent test of 4.6.8.

NOTE: If the auto-collimator contains an erecting system, the center of the target image will appear 20 mils below the reference line of sight of the auto-collimator. If it does not comtain an erecting system, the center of the target will appear 20 mils above the reference line of sight of the auto-collimator.

4.6.4.2 Collimation, pre-vibration, infrared viewing area. - With the index pin “D” disengaged from the indes plate “F” and surface “C” of the leveling base level; view through the auto-collimator of the Support Assembly collimator 10548767 positioned at the infrared viewing area (Head Wedge 10514254) and observe the projected target image. The horizontal ant? vertical reticle lines of the auto-collimator shall be superimposed on the horizontal and vertical target lines within the tolerance specified in 3.7.2 as measured on the auto-collimator’s micrometer knobs. Record this collimation measurement for use in the subsequent test of 4.6.9.

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4.6.4.3 Collimation, unity power area. - Perform this test with the aid of the 26° Elevation. Calibration Standard that conforms to Drawing D10548766. The calibration standard shall be set-up in accordance with the set-up instructions specified on Drawing F10547987 and then positioned at the unity power area (Head Window 10514256). With the index pin “P” disengaged from the index plate “T” and surface “C” of the leveling base level, view through the eyepiece of the 3X Telescope 5549108, contained in the calibration standard, and observe the boresight point (intersecting point of the horizontal and vertical reticle lires) of the telescope with respect to the boresight point of the target image. The boresight point of the telescope reticle shall be within the tolerance specified in 3.7.3, as measured on the target.

4.6.4.4 Backlash. - This test shall be performed subsequent to the test in 4.6.4.3. With the index pin, "P" disengaged from the index plate “E”, place the 10 second level on surface “C" of the leveling base. With backlash removed for direction of travel, rotate the leveling base by adjusting the leveling rod (Part #62 depicted on Drawing F8247391) until the bubble of the precision level is centered within its graduation lines without overtravel. View through the auto-collimator of the Support Assembly Collimator 10548767 positioned at the daylight viewing area and position the horizontal and vertical reticle lines of the auto-collimator coincident with the horizontal and vertical target image lines by means of the auto-collimator's micrometer knobs. Continue rotation of the leveling base approximately 2° in the same direction of travel, then reverse rotation of the leveling base (backlash included) until the bubble of the precision level is re-centered within its graduation lines without overtravel. View through the auto-collimator and observe for coincidence of the auto-collimator’s horizontal reticle line with respect to the horizontal target image lines. Any adjustment of the auto-collimator’s micrometer knobs to re-obtain coincidence of the aut.o-collimator’s horizontal recircle line with respect to the horizontal target image line is the amount of backlash. Backlash shall not exceed the tolerance specified in 3.7. 7. Upon satisfaction completion of this test, disemgage the leveling rod from the slot of the leveling base, engage the index pin “D” in the zero degree position of the index plate “E” and perform the following tests of 4.6.4.5 through 4.6.4.9 inclusive.

NOTE: Should overtravel occur during the test for backlash, the test shall be nullified and the procedure specified for backlash shal1 be performed.

4.6.4.5 Excursion range. - With the input coupling positioned as specified in 3.6b and the index pin “P” engaged in the zero degree postion on the index plate “E”, view through the auto-collimator positioned at the daylight viewing area and, if necessary, position the horizontal and vertical reticle lines of the auto-collimator coincident with the horizontal and vertical target image lines by means of the auto-collimator's micrometer knobs. Whem coincidence has been established, view, through the auto-collimator to the reflector “H” on the test fixture. Observe the auto-collimator reticle and note the position of the boresight point

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(intersecting point of the horizontal and vertical reticle lines) with respect to its reflected image. Disengage the index pin from the zero degree elevation position on the index plate. While observing the boresight point of the real ant! reflected image, rotate eht input compling and engaged the index pin, first in the 22 degree elevation position and then in the 18 degree depression position of the index plate. While the index pin is engaged at each position (22°EL & 18°DP), the noted position of the boresight point with respect to its reflected image shall not deviate in depression or elevation respectively to determine conformance with 3.7.4 (see 4.6.4.7 and 4.6.4.8 when performing this test).

4.6.4.6 Coupling travel. Rotate the coupling to its maximum elevation, then to its maximum depression stop position. While the coupling is at each stop, back off the coupling for at least 1/4 rotation and then again return the coupling to the stop. Perform this minimum of 3 times. During these rotations, the coupling shall show no evidence of irregularity of movement (sponginess or friction) while approaching or reaching the stop. Contact with the stop shall be solid, such that a distinct metal click can be heard to verify that coupling rotation is being limited only by means of contact with the coupling stop pin and not by contact of other components (optical or mechanical) to be in compliance with 3.7.4.1.

4.6.4.7 Plumb travel .- This test may be performed at the same tire that the test of 4.6.4.5 is performed. While observing the boresight point (intersecting point of the horizontal and vertical reticle lines) of the real and reflected image, rotate the input coupling so that the line of sight is deflected from 22° above to l8° below the noted position of the boresight point established in 4.6.4.5. The boresight point of the real and reflected image shall not deviate beyond the applicable value specified in 3.7.5 for the positions indicated as measured on the collimator reticle (see 4.6.4.8 when performing this test).

4.6.4.8 Elevation accuracy. - Perform this test in conjunction with the test in 4.6.4.7 (plumb, travel). The horizontal reticle lines of the real and reflected imaqe shall not deviate beyond the applicable value specified in 3.7.6 for the positions indicated.

4.6.4.9 Image tilt. - This test shall be performed with the index pin “D” engaged in the zero degree position on the index plate “F”. View through the auto-collimator positioned at the Head Hedge 10541997 and position one extremity of the auto-collimator’s vertical reticle line coincident to the vertical target image line. The image of the vertical, target line shall be parallel to the auto-collimators vertical reticle line within the tolerance specified in 3.7.8, as measured or the target. Repeat the above procedure with the auto-collimator positioned at the Head Wedge 10514254. The image of the vertical target line shall be parallel to the auto-collimator’s vertical reticle line within the tolerance specified in 3.7.8, as measured on tile target.

4.6.5 Resolution daylight and infrared viewing areas.- The test equipment specified in 4.6.4 is not required for this test (4.6.5) and the test i n4.6.6. Position the head assembly on its side with side “L” of

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the input coupling positioned 22° 30' with respect to the four mounting pads. This test shall be performed utilizing an observation telescope of at least 20 power with an entrance pupil as specified in 3.8.1 and a resolving power chart on a wall target or located in a collimator. The resolving power chart shall represent the angular subtense for the seconds of arc specified. Construction and positioning of the resolving power chart shall comply with the "Resolution test" specified in MIL-O-13830 and shall contain for line sets as shown in figure I therein. Position the observation telescope at the Head Wedge 10541997. The focus of the telescope shall be established at one best focus setting for viewing of the detectable line structure and proper line count in all four meridians of the resolving power chart. Resolution shall meet the requirement specified in 3.8.1. repeat the above procedure with the observation telescope positioned at the Head Wedge 10514254. The focus of the telescope shall be established at one best focus setting for viewing of the detectable line structure and proper line count in all four meridians of the resolving power chart. Resolution shall meet the requirement specified in 3.8.1.

4.6.6 Resolution, unity power system. - Perform this test subsequent to 4.6.5, utilizing an observation telescope of at least 3 power with an entrance pupil as specified in 3.8.2 and a resolving power chart on a wall target representing the angular subtense for the seconds of arc specified. Construction and positioning of the resolving power chart shall be as specified in 4.6.5. Resolution shall be within the tolerance specified in 3.8.2, as determined by viewing through the observation telescope when positioned at the Head Window 10514256.

4.6.7 Vibration.- This test shall be performed on head assemblies procured as a spare part item. When the head assemblies are provided as part of a complete periscope, this vibration test will not be required when the complete periscope is to be vibrated. Standard vibrating equipment capable of providing the amplitude and frequency specified in 3.5 shall be utilized for the vibration test. Unless othewise specified, the test equipment shall be in accordance with the "Test Facilities" requirements of MIL-F-13926. The head assembly shall be clamped on the vibrating equipment by means of an adapter that conforms to Drawing FS575736, Vibration Adapter for Periscope #32. The vil ratio nmotion shall be in the vertical plane for the time duration specified in 3.9. During this test, the input coupling shall be imobilized to any position that shall prevent it from contacting the travel stops. Immediately following the vibration cycle, and prior to the examination for physical defects resulting from poor workmanship and naterials, the head assembly shall be subjected to the tests in 4.6.8 and 4.6.9. Caution shall be exercied during transfer of the head assembly. Subsequent to the tests in 4.6.8 and 4.6.9, the head assembly shall be thoroughly examined for physical defects resulting from poor workmanship and materials to determine compliance with 3.9, prior to being subjected to the remaining tests specified herein. A visual examination shall be made for missing or damaged parts for irregularities of movement, friction or looseness. An audible examination, shall also be made for the head assembly for loose internal optical and mechanical commonents. Any one of the possible defects defined above shall be caused to

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consider a head assembly defective. Head assemblies which have passed the above examinations shall be subjected to the remaining tests specified in 4.6.10 to 4.6.12 inclusive.

4.6.8 Collimation, post-vibration, daylight viewing area. - Perform this test utilizing the procedure in 4.6.4.l. Record the collimation measurement. Any difference between the recorded measurement obtained in this test and the recorded measurement obtained in 4.6.4.1 shall not exceed the tolerance specified in 3.10.

4.6.9 Collimation, post-vibration, infrared viewing area.- Perfor- this test utilizing the prodecure in 4.6.4.2. Record the collimation measurement. Any difference between the recorded measurement obtained in this test and the recorded measurment obtained in 4.6.4.2 shall not exceed the tolerance specified in 3.11.

4.6.10 Sealing. - This portion of the test shall be performed utilizing a pressure gage capable of reading between 0.25 to 0.50 pound per square inch gage (psig) and a dew point tester. The following test sequence shall be adhered to. Check th eintermal pressure of the head assembly utilizing the pressure gage specified herein. Then, test the dew point of the interior atmosphere of the head assembly by means of the dew point tester. Interior atmosphere of the head assembly shall ccnform tc the requirements of 3.12. Upon satisfactory commpletion of this portion of the test, internal pressurizing of the head assembly shall be performed with special testing equipment conforming to 4.5.2.3.1. Pressurize the head assembly with dry nitrogen gas, having a dew point at least as low as minus 25°F, to 5.0+ 0.10 psip in accordance with the procedure for internal pressure testing on Drawing F8565556. Retain the applied pressure in the head assembly for a minimum of 2 hours. During the first hour, the input coupling shall be rotated through one full excursion of it articulation range. At the end of the 2 hour period there shall be no evidence of leakage in excess of 0.2 psi as indicated on the pressure gage of the testing equipment, to determine compliance with 3.12. Upon satisfactory completion of the sealing test, the head assembly shall then be flused with the dry nitrogen gas specified herein. The interior atmosphere of the head assembly shall be periodically sampled through the dew point tester until the atmosphere shows a dew point at least as low as minus 25°F. When this atmosphere is attained within the head assembly, the head assembly shall then be sealed a thhe pressure specified in 3.12.

4.6.11 Cleanliness. - The optical surfaces of the head assembly shall be inspected for cleanliness with the unaided eye. While viewing through each end of the module, inspection for defects shall be make by the technique of shadowing. While viewing through each Head Wedge (10541997 and 10514254), inspection for dirt shall he made by viewing an illuminated field provided by 2 each 15 watt desk fluorescent lamps held close to the Head Window 10514255. The head assembly shall conform to the requirements of 3.1% and 3.13b.

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4.6.12 Interchangeability. - The cavity at the Head Wedge 8289187 shall be inspected utilizing the maximum and minimum interchangeability gages conforming to Drawings D11732107 and D11732108 respectively, to determine compliance with 3.2. Upon successful completion of this portion of the test, the cavity at the Head Wedge 10514254 shall be inspected utilizing the maximum and minimum interchaneability gages conforming to Drawings D11732101 and D11729723 respectively, to detemine overall compliance with 3.2.

5. PREPARE FOR DELIVERY

5.1 Packaging, packing and marking. - Packaging, packing and marking shall be in accordance with Packaging Data Sheet 1054199O. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. - The Head Assembly 10541990 is a modular unit of the Tank Gunner's Periscope M32C primarily used in the fire control system of the 165101 Gun, Combat Engineer Vehicle: M728.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Selection of applicable levels of Preservation, packaging and packing.
c. Applicable packaging data (see 5.1).
d. Applicable stock number.
e. Provisions for first article testing.

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Army - MU

Preparing Activity:

Army - MU

Project No. 1240-A530

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-H-50646A(MU) Head Assembly: 10541990 (Periscope: M32C)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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MIL-H-50646A(AR)
AMENDMENT 1
28 October 1983

MILITARY SPECIFICATION

HEAD ASSEMBLY: 10541990 (PERISCOPE: M32C)

This Amendment forms a part of Military Specification MIL-H-50646A (MU), dated 6 August 1973, and is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 4

3.12. Line 2: Delete “0.2 pounds per square inch (psi)” and substitute “0.1. pound per square inch (psi)”.

line 3: Delete “minimum Of 2 hours” and substitute “minimum of 1 hour”.

PAGE 14

4.6.10, line 2: Delete “0.25 to 0.50” and substitute “0.0 to 0.50”.

line 14: Delete “minimum of 2 hours” and substitute “minimum of 1 hour”.

line 16: Delete “2 hour period” and substitute “1 hour period”. Delete “in excess Of 0.2" and substitute “in excess of 0.1”.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1240 -A828)

FSC 1240

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